

Computer Science

Level #4

Chapter #1 & Chapter #2

Computer Vision

2021

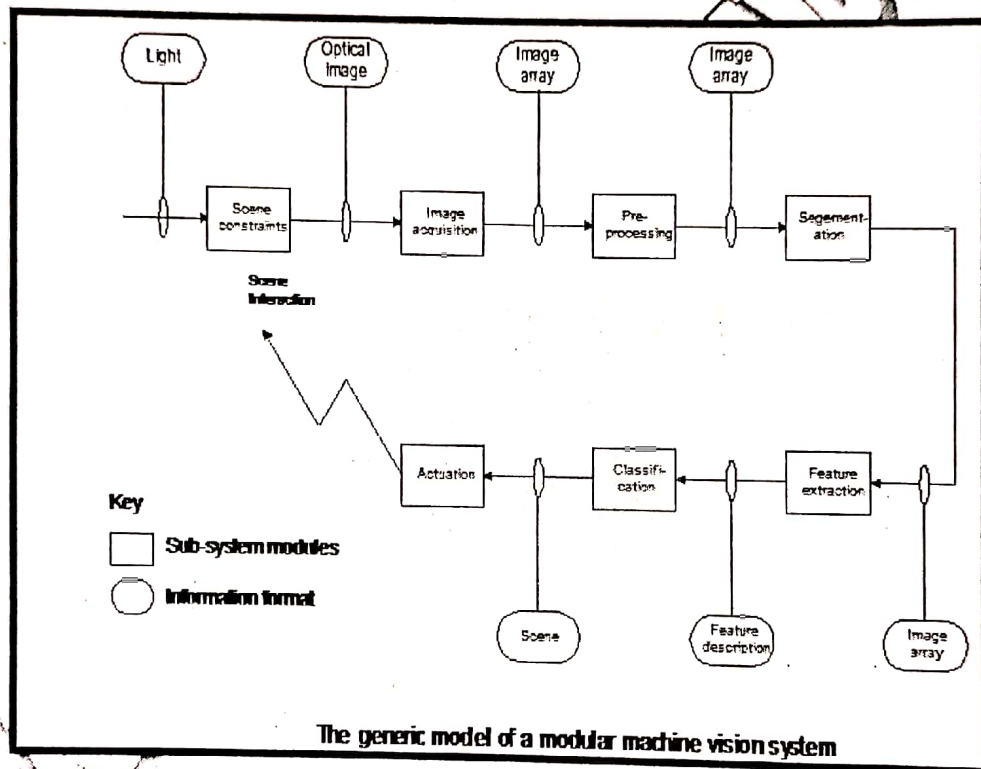
YOU ARE INVITED

Q1. What are the goal of any vision system?

⇒ The goal of any vision system Identify and locate a specific object in the scene.

Q2. What is the seven Subsystems of machine vision – Draw diagram?

- 1) Scene Constraints.
- 2) Image Acquisition.
- 3) Preprocessing.
- 4) Segmentation.
- 5) Feature Extraction.
- 6) Classification.
- 7) Actuation.



Q3. Describe the information available at each stage of generic machine vision system.

- 1) After the image acquired, the information is in array.
- 2) The processing and segmentation produce the same image format.
- 3) The feature extraction produce a list of feature descriptions.
- 4) The classification operation produce descriptions of the scene.

Q4. Describe the aim of scene constraints, and list some of these constraints.

⇒ The aim is encapsulate desired information in a form which easy to extract.

⇒ Achieving good control will result in:

- 1) Less complex.
- 2) More reliable.
- 3) Less expensive.

⇒ Constraints might include:

- 1) The number of objects.
- 2) Installation of cameras.
- 3) Special lighting conditions.

Chapter 2

Q5. What is the aim of the image acquisition sub-system?

⇒ An image is a projection of a 3D scene into a 2D projection plane.

⇒ Transform optical image data into an array of numerical data.

⇒ Each pixel defined as a 2 variable function $I(x, y)$.

Q6. List the three major issues for image acquisition.

- 1) Sensing.
- 2) Representation.
- 3) Digitization.

Q7. State the most commonly used devices for image sensing.

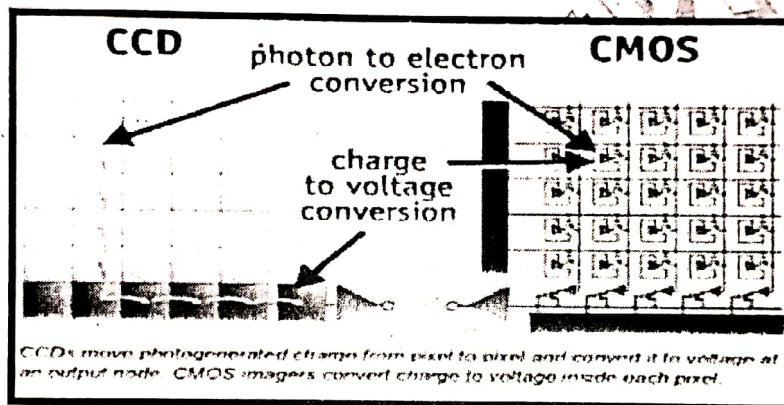
- 1) Vidicon cameras.
- 2) Solid-state Image Devices.
- 3) Laser scanners.

Q8. Explain briefly the Vidicon camera device

- ⇒ Device used to transform optical images into electrical signals.
- ⇒ An electron beam in the tube is deflected to scan the image.
- ⇒ The signal contains a large component of high-frequency noise.
- ⇒ Easily broken by vibration or shock.

Q9. Explain briefly the solid- state image devices.

- ⇒ These elements are a higher resolution.
- ⇒ The two main kinds of sensors used in digital cameras are:
 - 1) CCD (Charge coupled device).
 - 2) CMOS (complementary metal oxide on silicon).



Q10. Identify the necessary requirements for representation of image data.

- 1) Processing by means of a computer.
- 2) Contain all the information of the image.

Q11. Explain the different forms of image quantization.

⇒ **Sampling: Spatial Quantization**

- 1) The image is sampled at $(m \times n)$ points.
- 2) Each sample is called a picture cell (pixel).
- 3) Pixels is the smallest addressable area of a display.
- 4) We refer to a pixel by the number of its row and column.
- 5) **Resolution** number of pixels in a digital image.

⇒ Amplitude Quantization

- 1) Each pixel assigned a numerical code.
- 2) The code represents the intensity at that point.
- 3) Quantization: divide the vertical axis into pieces.
- 4) 8 bit quantization divides the vertical axis into 256.(2^8)

Q12. What is digital image, and state its types.

⇒ Is a representation of a two-dimensional image as a set of digital values, called pixels.

1) Black & white images:

- ✓ Binary images (1-bit images)
- ✓ Grayscale images (8-bit gray-level images)

2) Color images:

- ✓ 24-bit color images.
- ✓ 8-bit color images.

Q13. Define each of the following.

⇒ **Pixels:** picture elements in digital images

⇒ **Image Resolution:** number of pixels in a digital image :

⇒ **Resolution** = width x height

⇒ **File size** = width x height x #ofBytesPerPixel.

Q14. Discuss binary images?

⇒ Each pixel is stored as a single bit (0 or 1).

⇒ Also called a 1-bit monochrome image since it contains no color.

⇒ Example for File size calculation: Resolution: 640 x 480

✓ File size = $640 \times 480 \times 1/8 = 38400$ bytes

Q15. How to obtain a binary from gray scale images?

⇒ To generate Binary image from grey scale image $f(x,y)$.

⇒ A Threshold value, T , is used to partition the image into pixels with just two values, such that :

✓ IF $f(x,y) \geq T$ THEN $g(x,y) = 1$

✓ IF $f(x,y) < T$ THEN $g(x,y) = 0$

Q16. Discuss gray-scale image.

⇒ Each pixel has a gray-value between 0 and 255.

⇒ The high values correspond to bright pixels and the low values correspond to dark pixels.

⇒ We use one byte of memory for each pixel.

⇒ The whole image is described by an array of numbers called matrix.

⇒ File size calculation: Resolution: 640×480

✓ File size = $640 \times 480 \times 1 = 307\,200$ Bytes
= 300 kB

Q17. Discuss color image.

⇒ The primary colors red, green and blue.

⇒ A color image is described by three matrices.

⇒ In the RGB color space, a color is represented by a triplet (R, G, B)

- 1) R red component.
- 2) G green component.
- 3) B blue component.

⇒ Pixels are represented by three numbers.

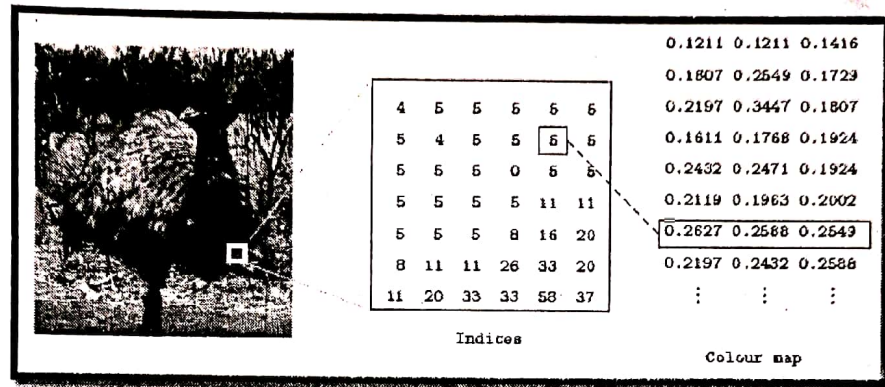
- 1) Red 0-255
- 2) Blue 0-255
- 3) Green 0-255

⇒ Each pixel is represented by three bytes.

✓ File size = $640 \times 480 \times 3 = 921\,600$ Bytes

Q18. Discuss indexed image.

- ⇒ One byte for each pixel.
- ⇒ Used in 8-bit color images is to store only the index, for each pixel.
- ⇒ Each code is actually an index into a table with 3-byte values that specify the color for a pixel with that lookup table index.
- ⇒ Look-Up Tables (LUTs) used to store color information.



Q19. Image of size 300 by 500 pixels, the dynamic range of gray levels is [0, 127]. Find the number of bits required to represent a pixel and size of the image in bytes.

1 2 4 7 16 32 64 128 256

- ⇒ Number of bits required to represent a pixel: 7 bits
- ⇒ Size of the image: $300 \times 500 \times 7/8 = 131250 \text{ Bytes} / 1024 = 128.173 \text{ KB}$.

Q20. The following image has integer intensities in the range between 0 and 63. Find the number of bits required to represent a pixel and size of the image in bytes.

0	0	9	4	1	2	2	2	2	2
2	6	5	8	6	2	40	4	62	63
2	8	7	3	2	4	10	3	60	40

1 2 4 7 16 32 64 128 256

- ⇒ Number of bits required to represent a pixel: 6 bits
- ⇒ Size of the image: $30 \times 10 \times 6/8 = 225 \text{ Bytes}$

Q21. Write Matlab code to read 'cells.jpg' image show it, size, rotate it 90 degree, show original and modified image.

```
>> im=imread('cells.jpg')
>> imshow(im)
>> size(im)
>> im2=imrotate(im,90)
>> imshow(im2)
>> imshow(im), figure, imshow(im2)
```

Q22. Perform the following operations using Matlab, type the results

⇒ Enter the following matrix:

a = [4, -2, -4, 7; 1, 5, -3, 2; 6, -8, -5, -6; -7, 3, 0, 1]

⇒ What is the result of these next commands

a(3,2) >> -8

a(8) >> 3

a(2, 1:3) >> [1 5 -3]

a(2:4, 3) >> $\begin{bmatrix} -3 \\ -5 \\ 0 \end{bmatrix}$

2.*a

$$\begin{pmatrix} 4 & -2 & -4 & 7 \\ 1 & 5 & -3 & 2 \\ 6 & -8 & -5 & -6 \\ -7 & 3 & 0 & 1 \end{pmatrix}$$